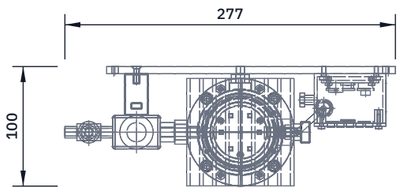
CONCEPT, NOT FOR FABRICATION

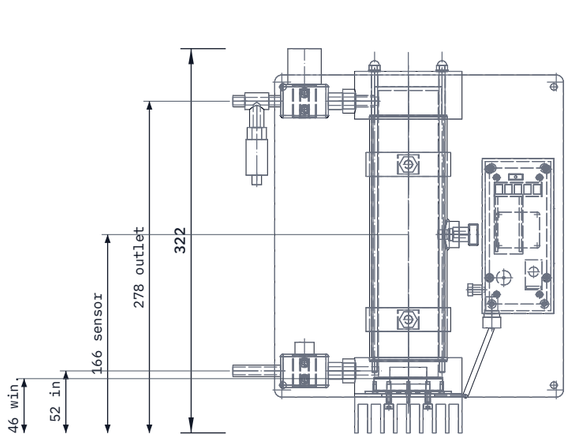
LumaFlow · github.com/BoujeeEnjinia1701/lumaflow

| REV | DESCRIPTION | DATE | BY |
|---|---|---|---|
| P1 | Preliminary GA for TRL 3 (from cad/src/model.py) | 2026-09-25 | AC |
| P2 | 50 mm bore, wall dose sensor, M5 rods (LMF-DDR-002) | 2026-09-25 | AC |
| P3 | Layout and labels tidied | 2026-09-30 | AC |
| P4 | Design for construction (LMF-DDR-003) | 2026-10-01 | AC |
| P5 | PTFE window washer, LED head plug, UV level bar, labels | 2026-10-02 | AC |
| P6 | Kit 1.9 detailed model; ePTFE window gasket (LMF-DEC-001 D-S1) | 2026-10-25 | AC |



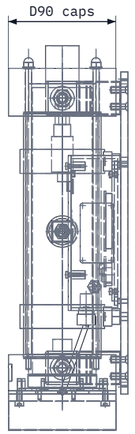

**TOP VIEW**

Scale 1:5; looking down (along -Z)



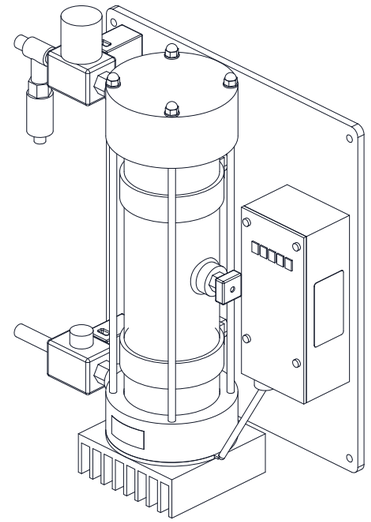

**FRONT VIEW**

Scale 1:5; looking at the front (along +Y)




**RIGHT VIEW**

Scale 1:5; looking at the right side (along -X)



**ISOMETRIC VIEW**

Not to scale; adapter not shown; seen from the front right and above, about 30 deg elevation

**MAIN DIMENSIONS AND INTERFACES (MM)**

Unit 277 W x 100 D x 322 H, adapter excluded
Channel D50 x 242; ports at Z 52 and 278
Dose sensor D8 window in tube wall at Z 166, facing +X
Tube D65 x 3 wall x 200, 316; high-refl. PTFE liner D59
Window D57 x 10 fused silica on a D51 seat
6 x 275 nm LEDs on a 32 mm circle, 1.3 mm below window
Caps D90 x 30 (316) and x 40 (acetal); 4 x M5 studs on D80
Ports 1/4 BSPP with push-fit stem adaptors; 316 outlet sleeve
Heat sink 90 x 90 x 30, fins down
Working pressure 8 bar; limiter at 4 bar upstream; flow up
Window from below on a 0.5 PTFE washer and 316 ring (M3 at 0.3 N m); ePTFE gasket
LED head plug with interlock loop under the enclosure; 5-segment UV level bar in the lid

LUMAFLOW

github.com/BoujeeEnjinia1701/lumaflow

# General arrangement

| DRAWING NO. | REV | SHEET | SIZE |
|---|---|---|---|
| LMF-DWG-001 | P6 | 1 of 1 | ANSI B |

| SCALE | UNITS | PROJECTION | DATE | DRAWN |
|---|---|---|---|---|
| 1:5 | mm | | 2026-10-25 | Amish Chadha |

MATERIAL / NOTES

316 tube and lower cap, high-reflectance PTFE liner, fused silica window, acetal upper cap; see bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION

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